

Simulation Problem (50 points):

Convert the design in HW3 to a fully-differential amplifier (only change is the load capacitor).

Considering load resistor $R_D=500\Omega$, the capacitor C of 20pF, and $V_{dd}=1.3V$, you need to design a 2-stage fully-differential cascaded common-source amplifier with a source follower driving 20pF.

You are to use an 130nm CMOS process. The design goals for this circuit are as follows:

Low frequency differential ac voltage gain > 12dB

Low-frequency common-mode AC gain > 80dB

Maximum output swing > 0.6Vpk-pk-diff (peak-to-peak-differential)

Unity gain bandwidth > 500MHz

What you need to turn in for this problem are:

- (15 pts) Hand calculation results including the values of DC bias voltage and current, transistor size, small-signal gain, output resistance, and maximum output swing.
- (15 pts) Cadence simulation results including a plot of small-signal gain and phase, and information showing DC conditions of the MOSFET.

Report: Attach schematics with DC operating points, AC gain and phase, transient simulation showing results at the bias points with the output and input swings. Note that you have to decide the right bias points to allow all the transistors M1, M2 and M3 to be in saturation. You also have to ensure that none of these transistors go out of saturation at the maximum or the minimum input voltages.

Changes from HW3:

- Your input source will have a differential AC components ($\pm 0.5V$) and a CM DC voltage (for $V_{GS1} + V_{\text{current-source, DS, sat}}$ in stage 1) as well as tail current source.
- Calculate both the differential and common-mode small-signal gain with the changes required.

